

A Simple Contrastive Embedding Framework for Low-Resource Fake News Detection

1 Appendix

1.1 Training

For all training and evaluation schemes, we employ NVIDIA DGX-A100 GPU with 40GB RAM and CUDA Version: 12.2. The hyperparameters are configured independently for each model. To ensure the reproducibility aspect, we configure our evaluation code to run deterministically by using a same set of random seeds across all models in this study. Table 1 shows the hyperparameters configured during training stage for all BERT models in this study.

Table 1: Hyperparameters for training BERT models in this study.

Hyperparameters	Zero-shot	MLM	E2E BERT	Pairwise	Triplet	Unsupervised
		Finetuning	Classifier	Contrast-BERT	Contrast-BERT	Contrast-BERT
BertConfig						
- hidden_act	'gelu'	'gelu'	'gelu'	'gelu'	'gelu'	'gelu'
- hidden_dropout_prob	0.1	0.1	0.1	0.1	0.1	0.1
- attention_probs_dropout_prob	0.1	0.1	0.1	0.1	0.1	0.1
- layer_norm_eps	1e-12	1e-12	1e-12	1e-12	1e-12	1e-12
Epoch	N/A	3	10	3	3	3
Learning rate	N/A	5e-5	2e-5	5e-5	5e-5	3e-5
Batch size	N/A	16	16	128	128	128/32
Max sequence length	N/A	320	160/512	160	160	160/512
mlm_prob	N/A	0.15	N/A	N/A	N/A	N/A

Table 2: Hyperparameters for training evidence-aware approaches on our Indonesian dataset.

Hyperparameters	MAC	GET	GETRAL	ReRead
Epoch	100	100	100	100
Learning rate	1e-5	1e-5	1e-5	1e-5
Batch size	32	32	32	
Max claim length	64	64	64	
Max evidence length	256	256	256	
use_claim_source	✗	✗	✗	N/A
use_article_source	✓	✓	✓	N/A
num_att_heads_for_words	2	2	2	N/A
num_att_heads_for_evds	1	1	1	N/A
gnn_window_size	N/A	3	3	N/A
gnn_dropout	N/A	0.8	0.8	N/A
gsl_rate	N/A	0.8	0.8	N/A
kernel_number	N/A	N/A	5	N/A
alpha	N/A	N/A	0.5	N/A
tau	N/A	N/A	0.07	N/A
dropout_left	0.8	0.8	0.8	
dropout_right	0.8	0.8	0.8	
decay_step	50	50	50	
decay_weight	0.01	0.01	0.01	